1. European Company AAA buys Crude Oil. Hedge Price Risk & Currency Risk.
Schedule of shipments:

	1M	2M	3M
Shipment of Crude Oil	1,000,000 Barrels	1,000,000 Barrels	1,000,000 Barrels

As a buyer we want to spend less money on purchasing oil. Final price equation:

$$Price = \frac{PriceUSD}{EUR/USD}$$

Best case $PriceUSD$ - ↓; EUR/USD - ↑. (Oil price decline + Weak dollar against the euro)

$$Price = \frac{PriceUSD - ProfitOIL}{EUR/USD + ProfitCUR}$$

FU($PriceUSD$) + FU(EUR/USD):

$$\text{M1: } Price = \frac{72.990}{1.1815} = 61.77$$

$$\text{M2: } Price = \frac{72.920}{1.1823} = 61.67$$

$$\text{M3: } Price = \frac{72.260}{1.1829} = 61.08$$

For crude oil price Barrel/USD:

- Speculation with CO (to buy) + PO (to buy) + FU (to sell)

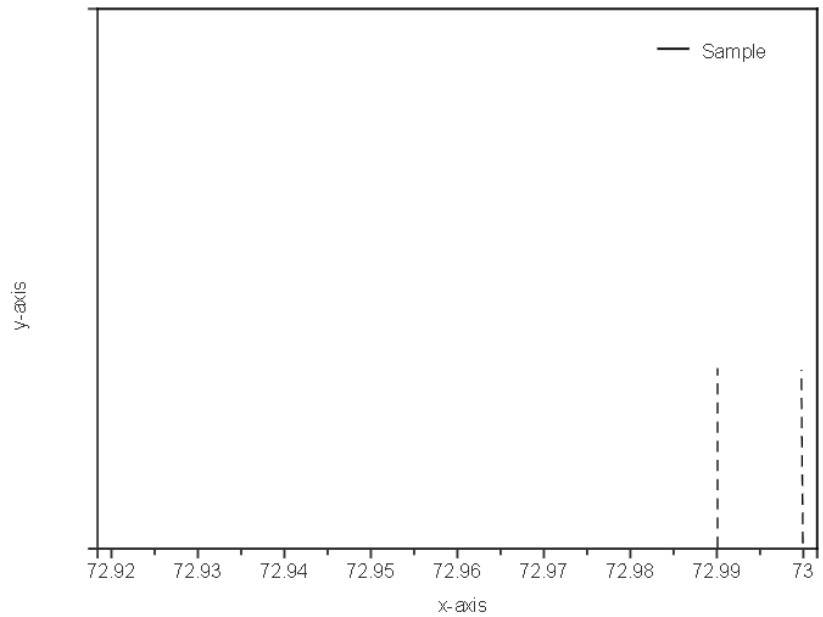
Product / Term	Spot	FU 1M	FU 2M	FU 3M
Crude Oil	72.920	72.990	72.920	72.260
Gasoline	2.150	2.282	2.154	2.093

Crude Oil

Put Option			Strike	Call Option		
3M	2M	1M		1M	2M	3M
1.60	1.20	0.50	65.000	10.50	11.80	13.50
2.40	2.10	1.30	67.000	8.40	9.60	11.30
3.90	3.30	2.60	69.000	5.20	7.60	7.80
8.20	7.20	5.30	71.000	3.30	3.70	4.70
11.40	9.60	8.70	73.000	2.10	2.20	2.60
13.30	12.10	10.40	75.000	0.60	0.90	1.70

M1

CO (to buy) Right / No obl. to buy | **Strike II** (73) | **Cost I** (2.1)
PO (to buy) No Right / Obl. to buy | **Strike II** (73) | **Cost II** (8.7)
FU (to sell) Right / Obl. to sell | **Strike I** (72.990)



1 MONTH:

- **Spot (72.920) < Strike I (72.990):**

  + 8.7
  - 2.1 } premium

  - 0.0
  + 0.0 } CO don't work

  - 73.0
  +72.920 } PO work

  +72.990
  - 72.920 } FU
  + 6.590

  Cost II - Cost I + Strike I(FU) - Strike II = +6.590$

- **Strike I (72.990) < Spot (72.995) < Strike II (73):**

  + 8.7
  - 2.1 } premium

  - 0.0
  + 0.0 } CO don't work

  - 73.0
  +72.995 } PO work

  +72.990
  - 72.995 } FU
  + 6.590

  Cost II - Cost I + Strike I(FU) - Strike II = +6.59$

- **Strike II (73) < Spot (75):**

  + 8.7
  - 2.1 } premium

  - 73.0
  + 75.0 } CO work

  - 0.0
  + 0.0 } PO don't work

  +72.990
  - 75.000 } FU
  + 6.590

  Cost II - Cost I + Strike I(FU) - Strike II = +6.59$

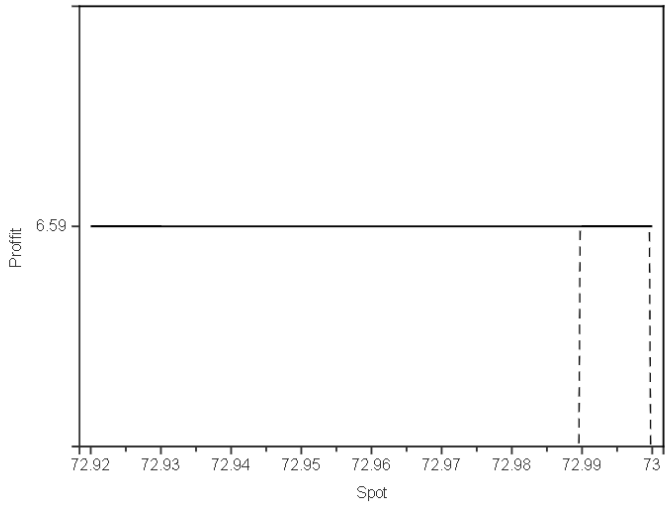


Figure1 - M1 Contract

M1
Cost II - Cost I + Strike I(FU) - Strike II = 9.6 - 2.2 + 72.920 - 73.0 = **+6.590$**

M2
Cost II - Cost I + Strike I(FU) - Strike II = 9.6 - 2.2 + 72.920 - 73.0 = +7.320$

M3
Cost II - Cost I + Strike I(FU) - Strike II = 11.40 - 2.60 + 72.260 - 73.0 = +8.060$

Price we spend to **M1** contract **USD/bar**: **Spot - profit = Spot - 6.590$**
Price we spend to **M2** contract **USD/bar**: **Spot - profit = Spot - 7.320$**
Price we spend to **M3** contract **USD/bar**: **Spot - profit = Spot - 8.060$**

Advantages:

- No risk involved.
- Guaranteed (fixed) profit.
- Better results compared to a simple **Future(to buy) + Future(to sell)** strategy.

Disadvantages:

- Profit does not increase even if market volatility is high.

To prevent volatility of currency EUR/USD:

- Speculation with COI (to sell) + COII (to buy)

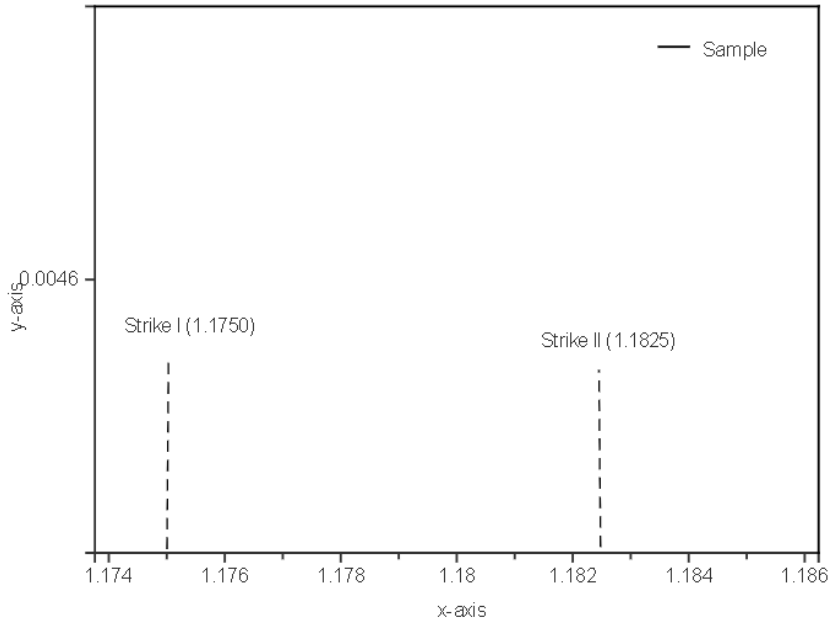
Currency	Spot	FU 1M	FU 2M	FU 3M
EUR/USD	1.1808	1.1815	1.1823	1.1829

Put Option			Strike	Call Option		
3M	2M	1M		1M	2M	3M
0.0060	0.0043	0.0033	1.1750	0.0117	0.0184	0.0207
0.0077	0.0059	0.0041	1.1775	0.0100	0.0150	0.0174
0.0097	0.0078	0.0051	1.1800	0.0085	0.0119	0.0144
0.0120	0.0102	0.0062	1.1825	0.0071	0.0093	0.011
0.0147	0.0130	0.0075	1.1850	0.0059	0.0071	0.0094
0.0178	0.0162	0.0089	1.1875	0.0048	0.0053	0.0075
0.0211	0.0198	0.0105	1.1900	0.0039	0.0039	0.0059

M1

COI (to sell) No Right / Obl. to sell **Strike I**(1.1750) | **Cost I** (0.0117)

COII (to buy) Right / No Obl. to buy **Strike II**(1.1825) | **Cost II** (0.0071)



1 MONTH:

- Spot (1.1740) < Strike I (1.1750):

  + 0.0117
  - 0.0071 } premium

  - 0.0
  + 0.0 } COI don't work

  + 0.0
  - 0.0 } COII don't work

  + 0.0046

  - **Profit** = Cost I - Cost II = +0.0046

- Strike I (1.1750) < Spot (1.1808) < Strike II (1.1825):

  + 0.0117
  - 0.0071 } premium

  + 1.1750
  - 1.1808 } COI work

  + 0.0
  - 0.0 } COII don't work

  - 0.0012

  - **Profit** = Cost I - Cost II + Strike I - Spot
    - if Spot = Strike I + Cost I - Cost II -> Profit 0
    - If Spot > 1.1796 -> lose on currency

- Strike II (1.1825) < Spot (1.1840):

  + 0.0117
  - 0.0071 } premium

  + 1.1750
  - 1.1850 } COI work

  - 1.1825
  + 1.1850 } COII work

  - 0.0029

- **Profit** = Cost I - Cost II + Strike I - Strike II = -0.0029
    - If Spot > 1.1825 -> max lose on currency

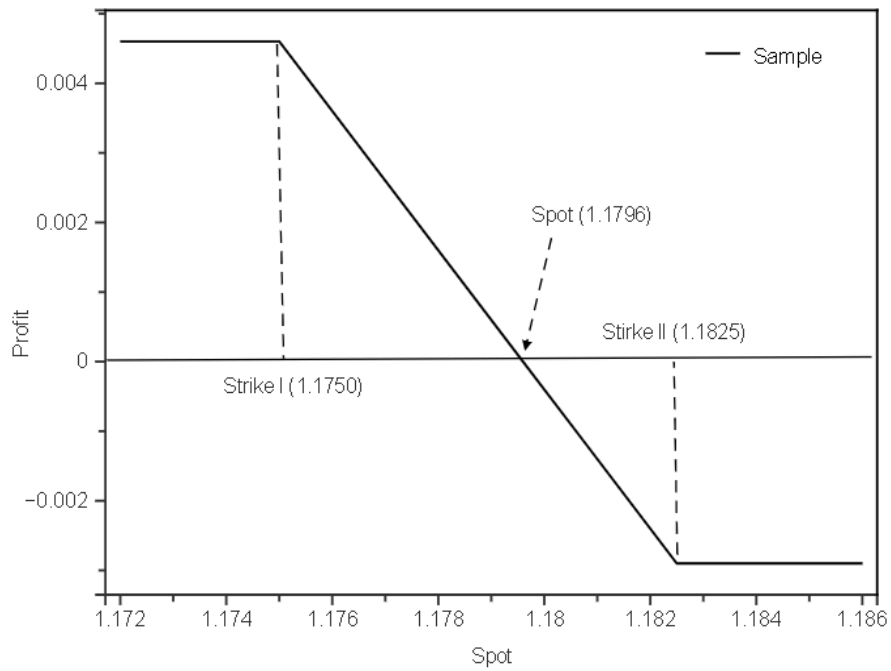


Figure 2 - 1M contract

2 MONTHS: (Arbitrage situation)

- Spot (1.1740) < Strike I (1.1750):
    - **Profit** = Cost I - Cost II = 0.0184 - 0.0093 =+ 0.0091
        - Max profit situation

- Strike I (1.1750) < Spot (1.1808) < Strike II (1.1825):
    - **Profit** = Cost I - Cost II + Strike I - Spot = + 0.0033

- Strike II (1.1825) < Spot (1.1840):
    - **Profit** = Cost I - Cost II + Strike I - Strike II = + 0.0016
        - Min profit situation

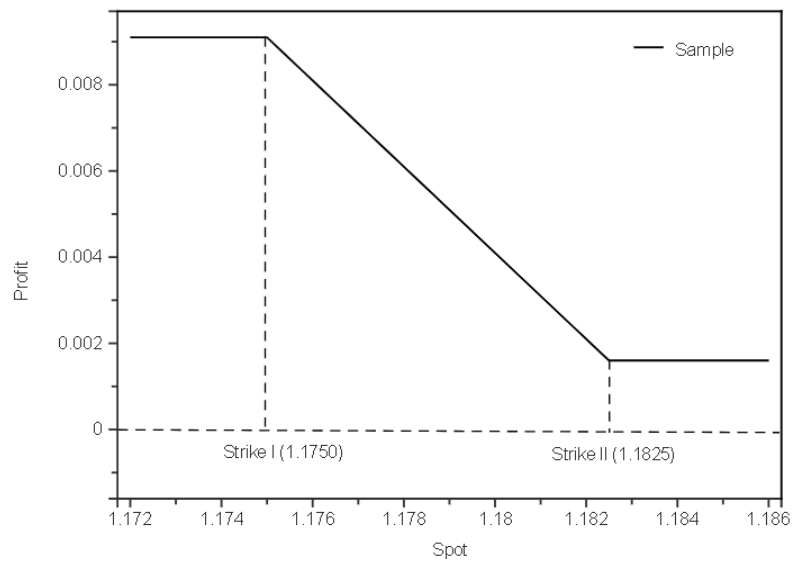


Figure 3 - 2M contract

3 MONTHS: (Arbitrage situation)

- Spot (1.1740) < Strike I (1.1750):
  - **Profit** = Cost I - Cost II = 0.0207 - 0.011 =+ 0.0093
    - Max profit situation

- Strike I (1.1750) < Spot (1.1808) < Strike II (1.1825):
  - **Profit** = Cost I - Cost II + Strike I - Spot = + 0.0039

- Strike II (1.1825) < Spot (1.1840):
  - **Profit** = Cost I - Cost II + Strike I - Strike II = + 0.0022
    - Min profit situation

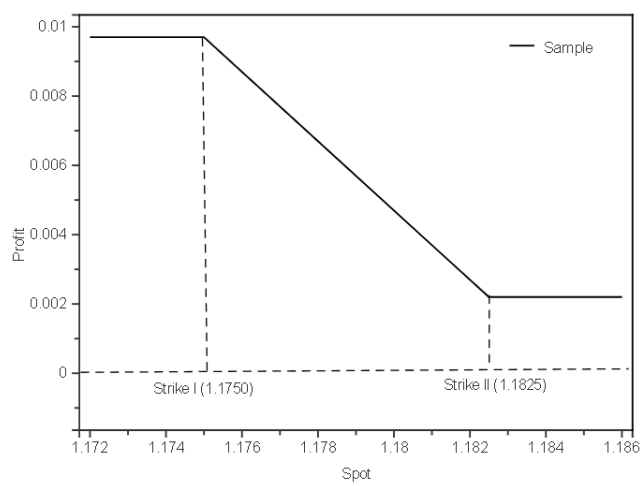


Figure 4 - 3M contract

- Spot < Strike I (1.1750) :
  - Price to **M1** contract **EUR/USD**: **Spot + profit = Spot** + 0.0046
  - Price to **M2** contract **EUR/USD**: **Spot + profit = Spot** + 0.0091
  - Price to **M3** contract **EUR/USD**: **Spot + profit = Spot** + 0.0093

- Strike I (1.1750) < Spot (1.1808) < Strike II (1.1825) :
  - Price to **M1** contract **EUR/USD**: **Spot + profit = Spot -** 0.0012
  - Price to **M2** contract **EUR/USD**: **Spot + profit = Spot** + 0.0033
  - Price to **M3** contract **EUR/USD**: **Spot + profit = Spot** + 0.0039

- Strike II (1.1825) < Spot :
  - Price to **M1** contract **EUR/USD**: **Spot + profit = Spot -** 0.0029
  - Price to **M2** contract **EUR/USD**: **Spot + profit = Spot** + 0.0016
  - Price to **M3** contract **EUR/USD**: **Spot + profit = Spot** + 0.0022

$$P_{final} = \frac{SPOToil - PROFIToil}{SPOTcurrency + PROFITcurrency}$$

Product / Term	Spot	FU 1M	FU 2M	FU 3M
Crude Oil	72.920	72.990	72.920	72.260

Currency	Spot	FU 1M	FU 2M	FU 3M
EUR/USD	1.1808	1.1815	1.1823	1.1829

M1	M2	M3
$Spot_{oil}$ = 72.920 ; $Spot_{currency}$ = 1.1808		
56.23 EUR/bar	55,40 EUR/bar	54,74 EUR/bar

WORST CASE(The highest price we will pay when …):

- $Spot_{currency}$ < Strike I (1.1750)
- $Spot_{oil}$ > Strike II (73.00)

M1	M2	M3
$Spot_{oil}$ = 75 ; $Spot_{currency}$ = 1.1740		
58,04 EUR/bar	57,20 EUR/bar	56,57 EUR/bar

BEST CASE(The Lowest price we will pay when …):

- $Spot_{currency}$ > Strike II (1.1825)
- $Spot_{oil}$ < Strike I (72.990)

M1	M2	M3
$Spot_{oil}$ = 72 ; $Spot_{currency}$ = 1.1830		
55.42 EUR/bar	54.60 EUR/bar	53.94 EUR/bar

FU($PriceUSD$) + FU(EUR/USD):

M1	M2	M3
61.77 EUR/bar	61.67 EUR/bar	61.08 EUR/bar

M1	M2	M3
$Spot_{oil}$ = 72 ; $Spot_{currency}$ = 1.1740		
55,4 EUR/bar	54.66 EUR/bar	54.03 EUR/bar
$Spot_{oil}$ = 72 ; $Spot_{currency}$ = 1.1808		
55.45 EUR/bar	54.62 EUR/bar	53.97 EUR/bar
$Spot_{oil}$ = 72 ; $Spot_{currency}$ = 1.1830		
55.42 EUR/bar	54.60 EUR/bar	53.94 EUR/bar

M1	M2	M3
$Spot_{oil}$ = 73 ; $Spot_{currency}$ = 1.1740		
56,34 EUR/bar	55,51 EUR/bar	54,88 EUR/bar
$Spot_{oil}$ = 73 ; $Spot_{currency}$ = 1.1808		
56,29 EUR/bar	55,46 EUR/bar	54,81 EUR/bar
$Spot_{oil}$ = 73 ; $Spot_{currency}$ = 1.1830		
56,27 EUR/bar	55,44 EUR/bar	54,7 EUR/bar

M1	M2	M3
$Spot_{oil}$ = 75 ; $Spot_{currency}$ = 1.1740		
58,04 EUR/bar	57,20 EUR/bar	56,57 EUR/bar
$Spot_{oil}$ = 75 ; $Spot_{currency}$ = 1.1808		
57,99 EUR/bar	57,15 EUR/bar	56,50 EUR/bar
$Spot_{oil}$ = 75 ; $Spot_{currency}$ = 1.1830		
57,96 EUR/bar	57,13 EUR/bar	56,47 EUR/bar